

Software Engineering Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

Project: Hospital Bed & ICU Allocation Dashboard

UTTAM B S

PES1UG24CS508

1. Objective

The objective of this lab was to convert functional requirements into Agile Epics and User Stories, prioritize and estimate the backlog, simulate three sprints in Jira, and study the sprint progress using Burndown Charts.

2. Epics

The project was divided into the following three major functional areas:

Epic 1: Real-Time Bed & ICU Monitoring – Enables hospital staff to monitor current bed and ICU availability.

Epic 2: Patient Triage & Priority Management – Enables hospital staff to record patient details and assign priority.

Epic 3: ICU Bed Allocation – Enables hospital staff to view, allocate, and update ICU bed status.

3. User Stories and Story Points

The original backlog contained eight User Stories. Story Points were assigned using Fibonacci-style values to represent relative effort and complexity. Five additional work items were later added for Sprint 3 during the Jira simulation.

Key	User Story	Epic	Priority	Points	Sprint
UTTAM-7	View Real-Time Bed Availability	Real-Time Bed & ICU Monitoring	High	1	Sprint 1
UTTAM-8	Monitor ICU Bed Availability	Real-Time Bed & ICU Monitoring	High	2	Sprint 1
UTTAM-9	Register Patient Details	Patient Triage & Priority Management	High	3	Sprint 1
UTTAM-13	Assign Patient Priority	Patient Triage & Priority Management	High	5	Sprint 1
UTTAM-10	View Available ICU Beds	ICU Bed Allocation	High	8	Sprint 2
UTTAM-11	Allocate ICU Bed to Patient	ICU Bed Allocation	High	13	Sprint 2
UTTAM-14	Update Bed	ICU Bed	Medium	21	Sprint 2

	Allocation Status	Allocation			
UTTAM-12	Update Bed Availability	Real-Time Bed & ICU Monitoring	Medium	34	Sprint 2

4. Jira Backlog

The backlog shows the eight User Stories, their Epic association, status, and Story Points.

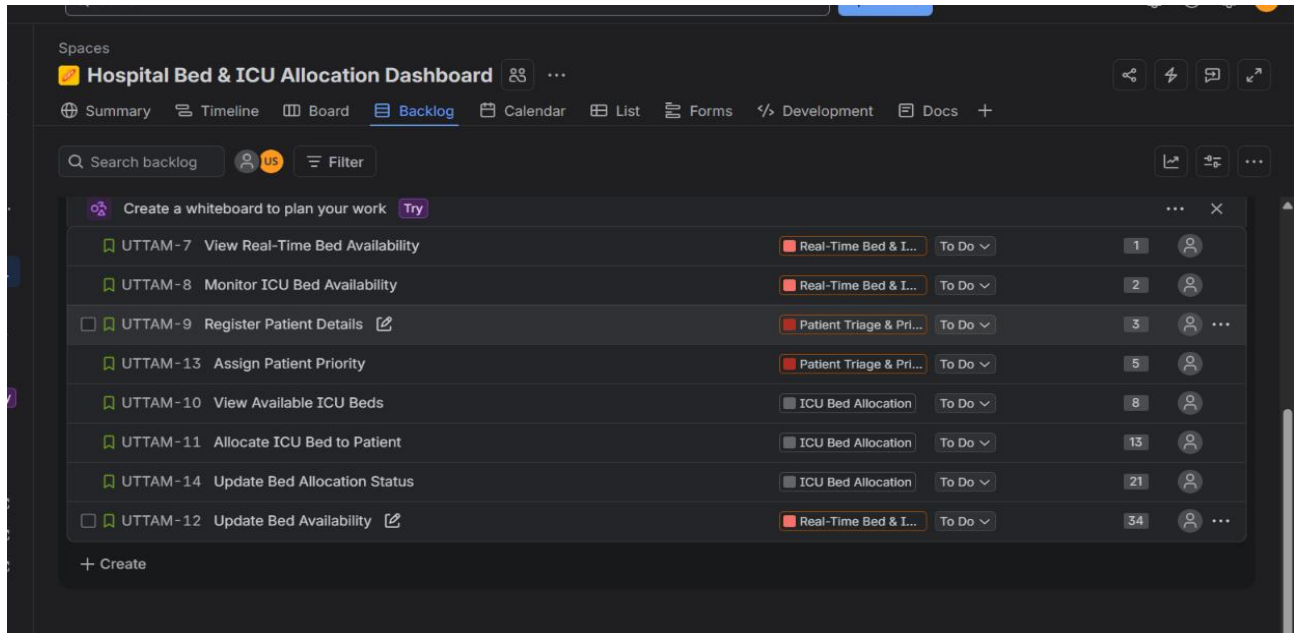


Figure 1: Jira backlog with User Stories and Story Points.

5. Sprint 1

Sprint 1 focused on the core monitoring and patient triage features. Four stories were included and completed.

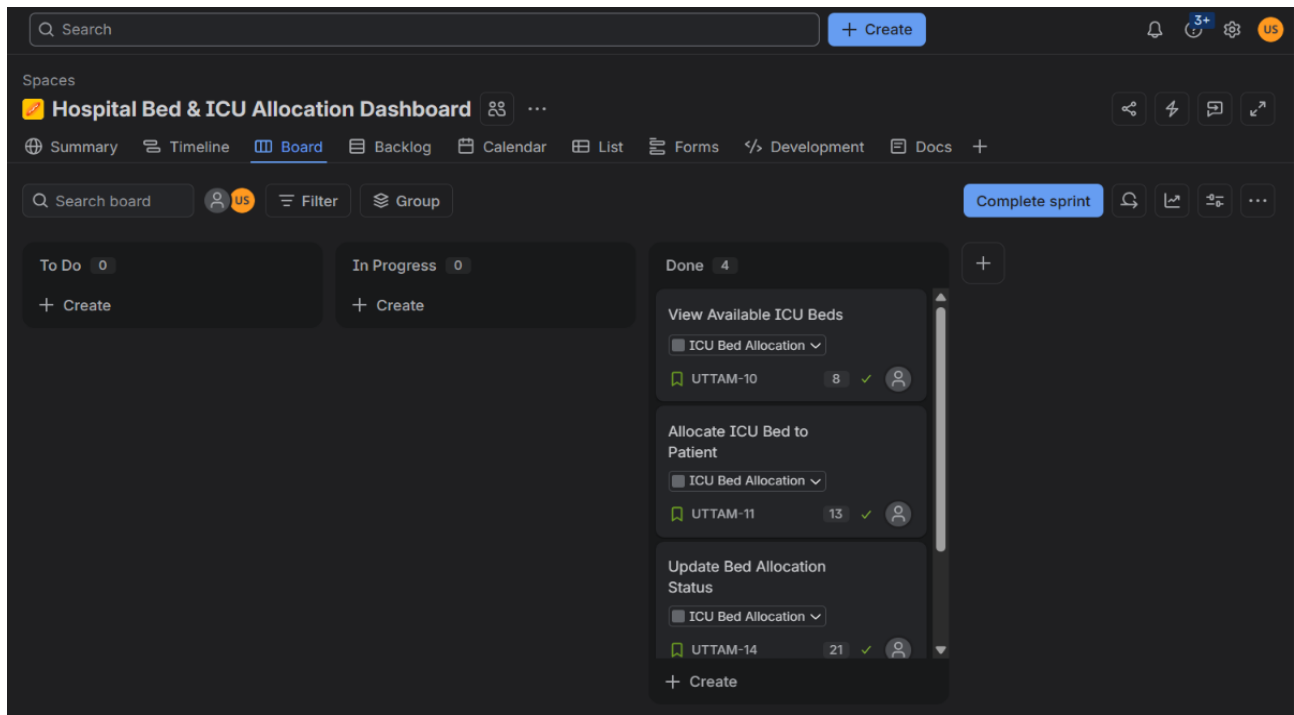


Figure 2: Sprint 1 board showing the completed work.

6. Sprint 2

Sprint 2 focused on ICU bed allocation and availability updates. The four selected stories were completed.

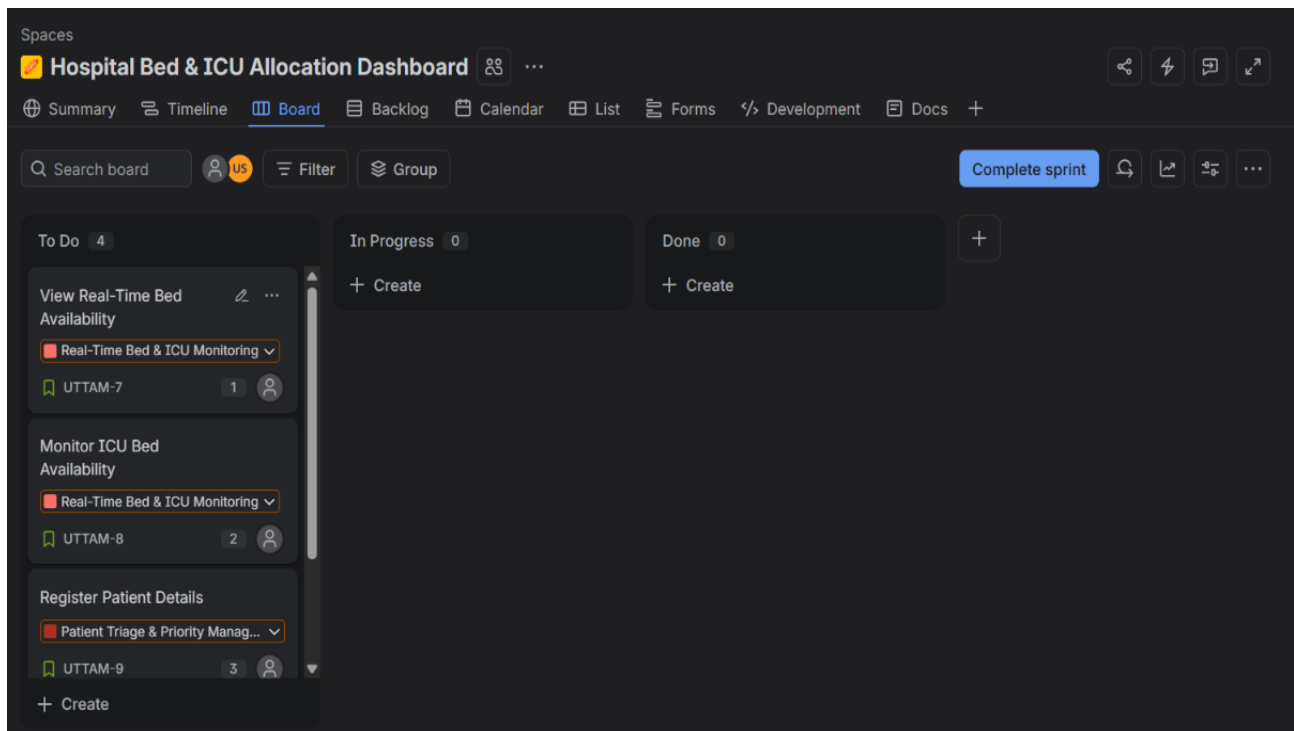


Figure 3: Sprint 2 board showing the completed work.

7. Sprint 3

Sprint 3 focused on additional bed monitoring, patient triage, and ICU bed allocation work. Five work items were included and completed, with a total of 19 Story Points.

Key	User Story	Epic	Status	Points
UTTAM-15	Search Beds by Availability	Real-Time Bed & ICU Monitoring	Done	1
UTTAM-16	View ICU Bed Status	Real-Time Bed & ICU Monitoring	Done	2
UTTAM-17	Update Patient Triage Information	Patient Triage & Priority Management	Done	3
UTTAM-18	View Patient Priority	Patient Triage & Priority Management	Done	5
UTTAM-19	Release Allocated ICU Bed	ICU Bed Allocation	Done	8

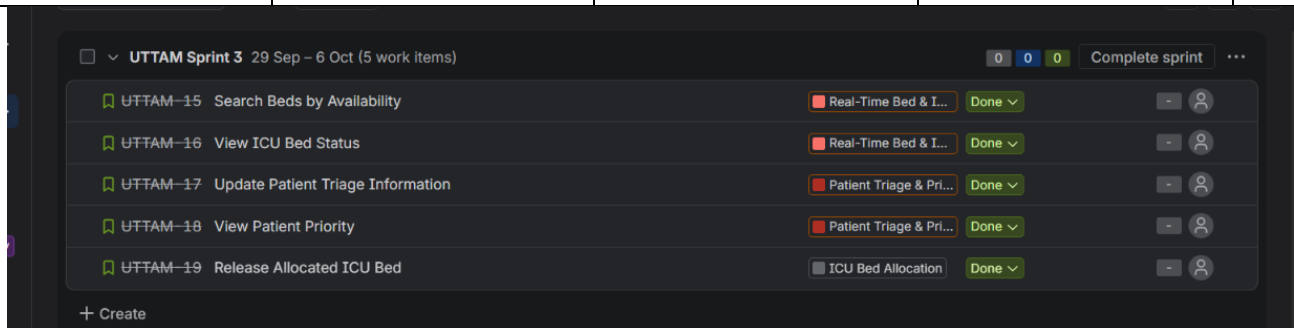


Figure 4: Sprint 3 board showing the completed work.

8. Burndown Charts

The Burndown Charts show the remaining work against the ideal burn-rate guideline for the three simulated sprints.

Sprint 1

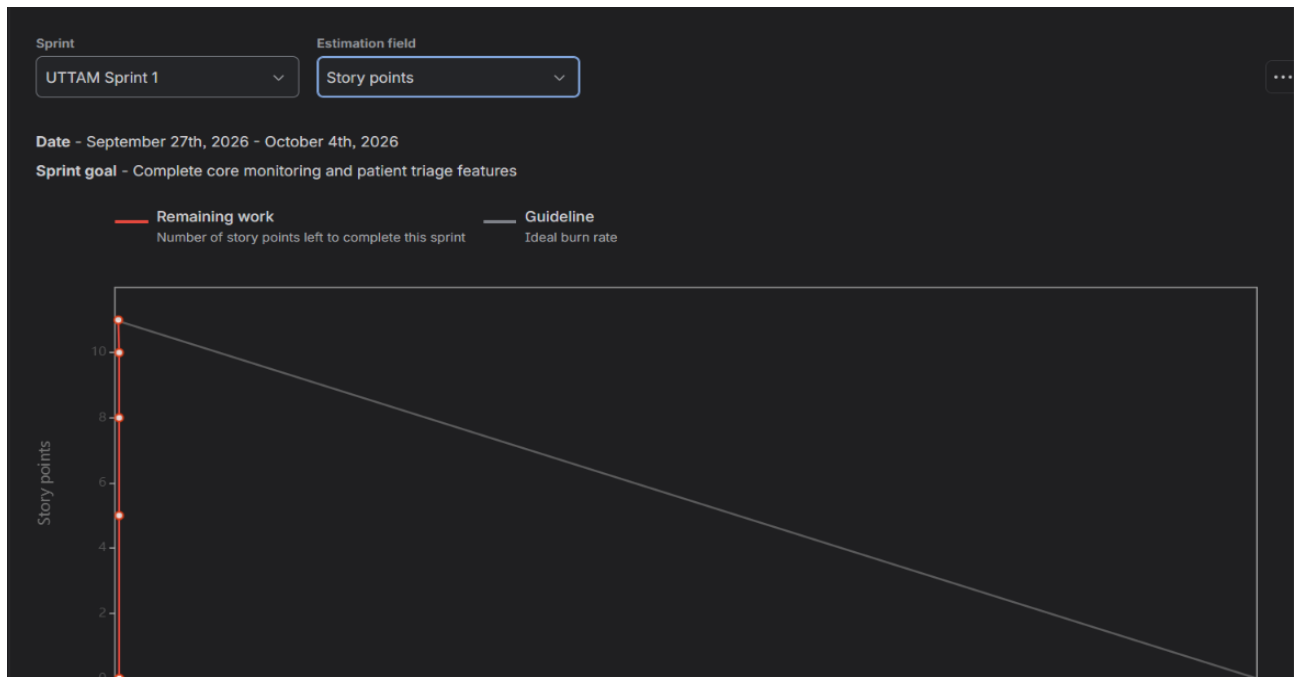


Figure 5: Sprint 1 Burndown Chart.

Sprint 2

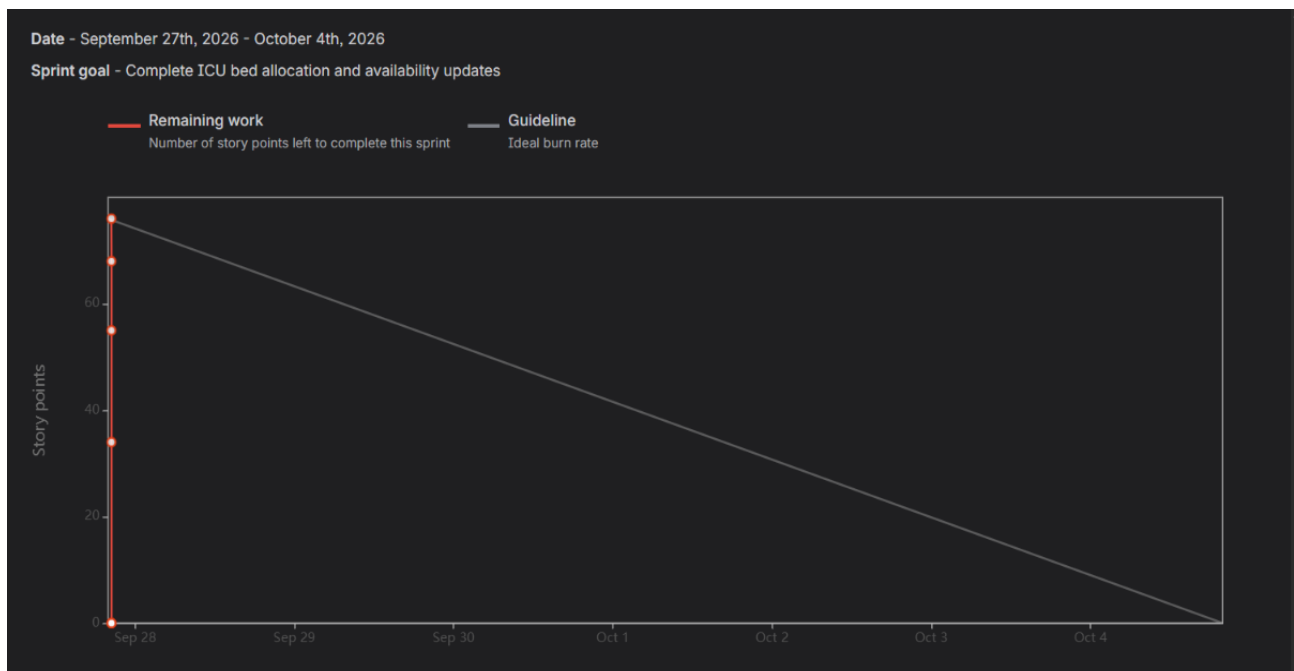


Figure 6: Sprint 2 Burndown Chart.

Sprint 3



Figure 7: Sprint 3 Burndown Chart.

9. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The story-point estimates gave a reasonable idea of the relative effort and complexity of the stories. The stories selected for the sprints were completed successfully.

2. Was your backlog well-prioritized?

Yes. The core functions such as bed monitoring, patient registration, patient priority, and ICU allocation were given High priority. Supporting update functions were given Medium priority.

3. How did your simulated sprint align with your plan?

The simulated sprints followed the planned order. Sprint 1 handled monitoring and triage, Sprint 2 handled ICU bed allocation and availability updates, and Sprint 3 added further monitoring, triage, and ICU allocation work. The selected stories were moved through the workflow and completed.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Charts showed the amount of remaining work and compared it with the ideal burn rate. They helped show how much work was planned for each sprint and whether the work was being completed within the available sprint period.

10. Conclusion

The Jira exercise demonstrated the basic Agile Scrum workflow. The project requirements were organized into Epics and User Stories, priorities and Story Points were assigned, three sprints were simulated, and Burndown Charts were used to review sprint progress.